

Mathematical Foundations for Machine Learning - Mid-Semester 1 Mock 01

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MFML Midsem 1 - Mock 01 Question Paper

Time: 90 minutes Maximum marks: 30 Scope: Sessions 1-8 Instructions: Attempt all questions. For every calculation-bearing response, show the formula, substitutions, non-obvious intermediate steps, shapes or domains, validity conditions, and interpretation. No reference material is permitted.

Q1 - Linear system classification (4 marks)

Sessions: 1 | Topics: consistency and uniqueness | Forms: numerical reasoning; scenario analysis or interpretation | Provenance: Authored BITS-style | Time guidance: 10 minutes Point difficulty mix: Easy 1, Standard 3, Stretch 0

An analyst models two real-valued calibration adjustments x and y by $x+y=3$ and $2x+2y=6$. Determine $\text{rank}(A)$ and $\text{rank}([A|b])$ and classify the solution set. Then decide whether the model determines a unique calibration, explain what the remaining freedom means, and describe what kind of additional equation would remove the ambiguity.

Q2 - Vector and affine spaces (4 marks)

Sessions: 2 | Topics: basis and affine spaces | Forms: property or definition; comparison or distinction | Provenance: Authored BITS-style | Time guidance: 9 minutes Point difficulty mix: Easy 2, Standard 2, Stretch 0

Define linear independence and spanning, then state how they combine to make a basis. State the zero, addition-closure, and scalar-closure requirements for a subspace U . Define the affine set $a+U$, compare its zero-vector requirement with that of U , and give the exact condition under which $a+U$ is itself a subspace.

Q3 - Projection (4 marks)

Sessions: 3 | Topics: norm, orthonormal direction, projection | Forms: numerical reasoning; scenario analysis or interpretation | Provenance: Authored BITS-style | Time guidance: 10 minutes Point difficulty mix: Easy 1, Standard 2, Stretch 1

Let $x=(3,4)$ and $q=(1/\sqrt{2}, 1/\sqrt{2})$. Verify that q is a unit vector, find the orthogonal projection of x onto $\text{span}(q)$, verify that the residual is orthogonal to q , and interpret the projection and residual geometrically.

Q4 - Eigenpairs and Cholesky (4 marks)

Sessions: 4 | Topics: eigenpairs and Cholesky | Forms: property or definition; numerical reasoning; scenario analysis or interpretation | Provenance: Authored BITS-style | Time guidance: 12 minutes Point difficulty mix: Easy 1, Standard 2, Stretch 1

State the defining equation and nonzero-vector condition for an eigenpair. Treat the matrix below as a covariance matrix, calculate a lower-triangular L with positive diagonal such that $A=LL^T$, and verify the product. Decide whether ordinary real Cholesky is valid and explain what that decision means for this covariance model.

$$A = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$$

Q5 - Diagonalization and SVD (4 marks)

Sessions: 5 | Topics: diagonalizability, invertibility, low-rank approximation | Forms: comparison or distinction; scenario analysis or interpretation | Provenance: Authored BITS-style | Time guidance: 11 minutes Point difficulty mix: Easy 1, Standard 2, Stretch 1

Decide whether each statement is true: (i) every real square matrix with distinct real eigenvalues is diagonalizable; (ii) every such matrix is invertible. Prove each true statement and construct a 2 by 2 counterexample for each false statement. A compression system replaces a real matrix with its best rank-2 approximation; the original singular values are 5,2,1. Calculate the spectral-norm error, then decide whether it meets an allowed error of at most 1.5 and interpret the decision.

Q6 - Jacobian interpretation (4 marks)

Sessions: 6 | Topics: partial derivatives and Jacobian | Forms: numerical reasoning; scenario analysis or interpretation | Provenance: Authored BITS-style | Time guidance: 12 minutes Point difficulty mix: Easy 1, Standard 2, Stretch 1

Use the output-by-input convention for $g: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, where $g(x,y)=(x^2+y,xy)$. Compute $J_g(2,3)$, state its shape, and use it to estimate the output change for input displacement $h=(0.01,-0.02)$. Interpret the two components as local changes.

Q7 - Backpropagation and automatic differentiation (3 marks)

Sessions: 7 | Topics: computation graph, reverse mode, AD | Forms: numerical reasoning; scenario analysis or interpretation | Provenance: Authored BITS-style | Time guidance: 12 minutes Point difficulty mix: Easy 1, Standard 1, Stretch 1

At $x=2$, a computation graph uses $u=x^2$, $y=3u+1$, $L=y^2$. Perform the forward and reverse passes to obtain dL/dx . Then explain how automatic differentiation differs from both symbolic differentiation and finite differences.

Q8 - Hessian and optimization conclusion (3 marks)

Sessions: 8 | Topics: stationary points, Hessian, local and global minima | Forms: numerical reasoning; scenario analysis or interpretation | Provenance: Authored BITS-style | Time guidance: 14 minutes Point difficulty mix: Easy 1, Standard 1, Stretch 1

For $f(x,y)=x^2+2y^2-4x+8y$, find the stationary point and classify it using the Hessian. Decide whether the minimum is local or global, justify the stronger conclusion, and report the minimum value.